

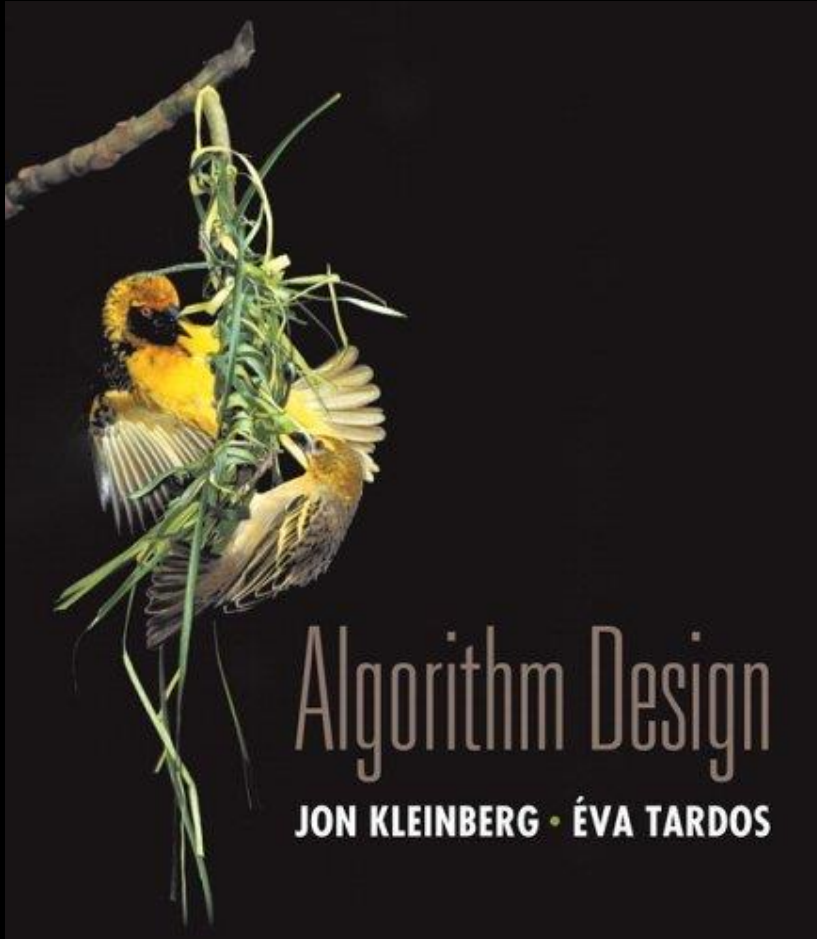
Course evaluation

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Chapter 12*

Local Search



Slides by Kevin Wayne.
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12.1 Landscape of an Optimization Problem

Hill-climbing: Vertex Cover

VERTEX-COVER. Given a graph $G = (V, E)$, find a subset of nodes S of minimal cardinality such that for each $u-v$ in E , either u or v (or both) are in S .

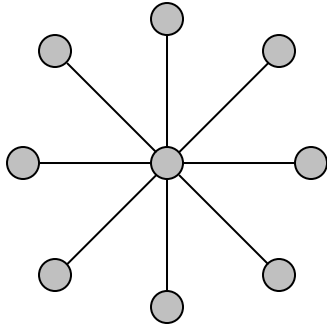
Neighbor relation. $S \sim S'$ if S' can be obtained from S by adding or deleting a single node. Each subset S has n neighbors.

Hill-climbing ("Gradient descent" in the textbook). Start with $S = V$. If there is a neighbor S' that is a vertex cover and has lower cardinality, replace S with S' .

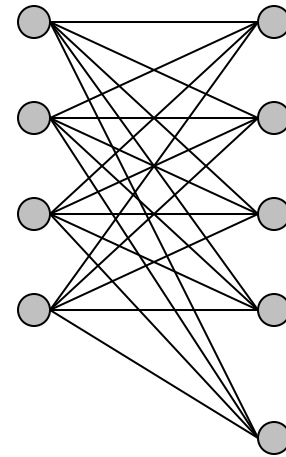
Remark. Algorithm terminates after at most n steps since each update decreases the size of the cover by one.

Hill-climbing: Vertex Cover

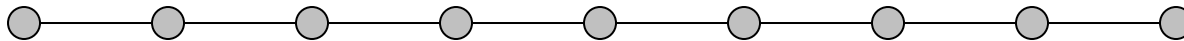
Local optimum. No neighbor is strictly better.



optimum = center node only
local optimum = all other nodes



optimum = all nodes on left side
local optimum = all nodes on right side



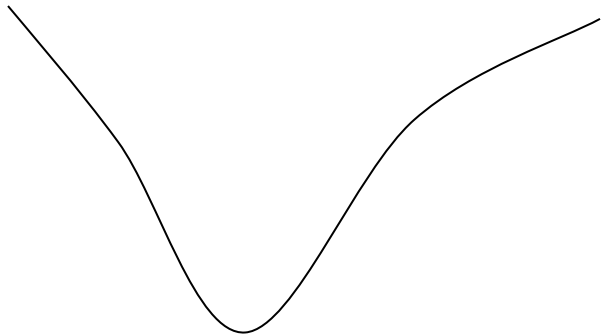
optimum = even nodes
local optimum = omit every third node

Local Search

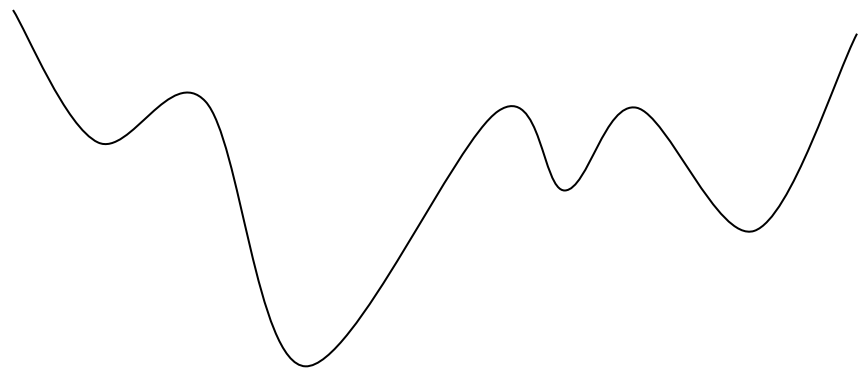
Local search. Algorithm that explores the space of possible solutions in sequential fashion, moving from a current solution to a "nearby" one.

Neighbor relation. Let $S \sim S'$ be a neighbor relation for the problem.

Hill-climbing. Let S denote current solution. If there is a neighbor S' of S with strictly lower cost, replace S with the neighbor whose cost is as small as possible. Otherwise, terminate the algorithm.



A funnel



A jagged funnel

12.4 Maximum Cut

Maximum Cut

Maximum cut. Given an undirected graph $G = (V, E)$ with positive integer edge weights w_e , find a node partition (A, B) such that the total weight of edges crossing the cut is maximized.

$$w(A, B) \coloneqq \sum_{u \in A, v \in B} w_{uv}$$

Toy application.

- n activities, m people.
- Each person wants to participate in two of the activities.
- Schedule each activity in the morning or afternoon to maximize number of people that can enjoy both activities.

Real applications. Circuit layout, statistical physics.

Maximum Cut

Single-flip neighborhood. Given a partition (A, B) , move one node from A to B , or one from B to A if it improves the solution.

Greedy algorithm.

```
Max-Cut-Local (G, w) {  
    Pick a random node partition (A, B)  
  
    while ( $\exists$  improving node v) {  
        if (v is in A) move v to B  
        else           move v to A  
    }  
  
    return (A, B)  
}
```

Maximum Cut: Local Search Analysis

Theorem. Let (A, B) be a locally optimal partition and let (A^*, B^*) be optimal partition. Then $w(A, B) \geq \frac{1}{2} \sum_e w_e \geq \frac{1}{2} w(A^*, B^*)$.

↑
weights are nonnegative

Pf.

- Local optimality implies that for all $u \in A$: $\sum_{v \in A} w_{uv} \leq \sum_{v \in B} w_{uv}$

Adding up all these inequalities yields:

$$2 \sum_{\{u,v\} \subseteq A} w_{uv} \leq \sum_{u \in A, v \in B} w_{uv} = w(A, B)$$

- Similarly $2 \sum_{\{u,v\} \subseteq B} w_{uv} \leq \sum_{u \in A, v \in B} w_{uv} = w(A, B)$

- Now,

each edge counted once

$$\sum_{e \in E} w_e = \underbrace{\sum_{\{u,v\} \subseteq A} w_{uv}}_{\leq \frac{1}{2} w(A, B)} + \underbrace{\sum_{u \in A, v \in B} w_{uv}}_{w(A, B)} + \underbrace{\sum_{\{u,v\} \subseteq B} w_{uv}}_{\leq \frac{1}{2} w(A, B)} \leq 2w(A, B) \quad \blacksquare$$

Maximum Cut: Big Improvement Flips

Local search. Within a factor of 2 for MAX-CUT, but not poly-time!

Big-improvement-flip algorithm. Only choose a node which, when flipped, increases the cut value by at least $\frac{2\varepsilon}{n} w(A, B)$

Claim. Upon termination, big-improvement-flip algorithm returns a cut (A, B) with $(2 + \varepsilon) w(A, B) \geq w(A^*, B^*)$.

Pf idea. Add $\frac{2\varepsilon}{n} w(A, B)$ to each inequality in original proof.

Claim. Big-improvement-flip algorithm terminates after $O(\varepsilon^{-1} n \log W)$ flips, where $W = \sum_e w_e$.

- Each flip improves cut value by at least a factor of $(1 + \varepsilon/n)$.
- After n/ε iterations the cut value improves by a factor of 2.
- Cut value can be doubled at most $\log W$ times.

↑
if $x \geq 1$, $(1 + 1/x)^x \geq 2$

Maximum Cut: Context

Theorem. [Sahni-Gonzales 1976] There exists a $\frac{1}{2}$ -approximation algorithm for MAX-CUT.

Theorem. [Goemans-Williamson 1995] There exists an 0.878567-approximation algorithm for MAX-CUT.

$$\min_{0 \leq \theta \leq \pi} \frac{2}{\pi} \frac{\theta}{1 - \cos \theta}$$

Theorem. [Håstad 1997] Unless $P = NP$, no $16/17$ approximation algorithm for MAX-CUT.

$$\uparrow$$

0.941176

Chapter Summary

Local Search

Local search. Algorithm that explores the space of possible solutions in sequential fashion, moving from a current solution to a "nearby" one.

Algorithms

- Hill-climbing

Problems

- Max-cut